



US012408288B2

(12) **United States Patent**
Chianghsieh et al.

(10) **Patent No.:** **US 12,408,288 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **RISER CAGE HALF TURN FASTENER**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 174 days.

(21) Appl. No.: **18/489,257**

(22) Filed: **Oct. 18, 2023**

(65) **Prior Publication Data**

US 2025/0133679 A1 Apr. 24, 2025

(51) **Int. Cl.**
F16B 21/09 (2006.01)
H05K 7/14 (2006.01)

(52) **U.S. Cl.**
CPC **H05K 7/1487** (2013.01); **F16B 21/09**
(2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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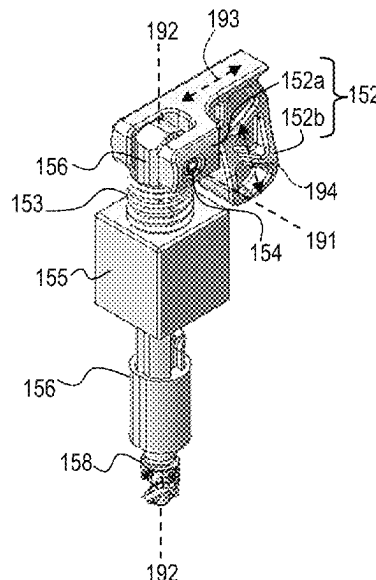
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(57) **ABSTRACT**

A riser cage of a computing device comprises a bracket, a riser card attached to the bracket, and a fastener attached to the bracket. The bracket is to support an expansion card. The riser card is to electrically connect the expansion card to the computing device. The fastener is to secure the riser cage to the computing device. The fastener comprises a handle, a shaft extending along a first axis, a pivot connecting the handle to the shaft, and a half-turn latch coupled to the shaft. The half-turn latch is configured to engage with a receptacle of the computing device to selectively secure the fastener to the receptacle. The handle is pivotable about the pivot between a deployed configuration and a stowed configuration, and in the stowed configuration the handle is parallel to the riser card.

20 Claims, 7 Drawing Sheets



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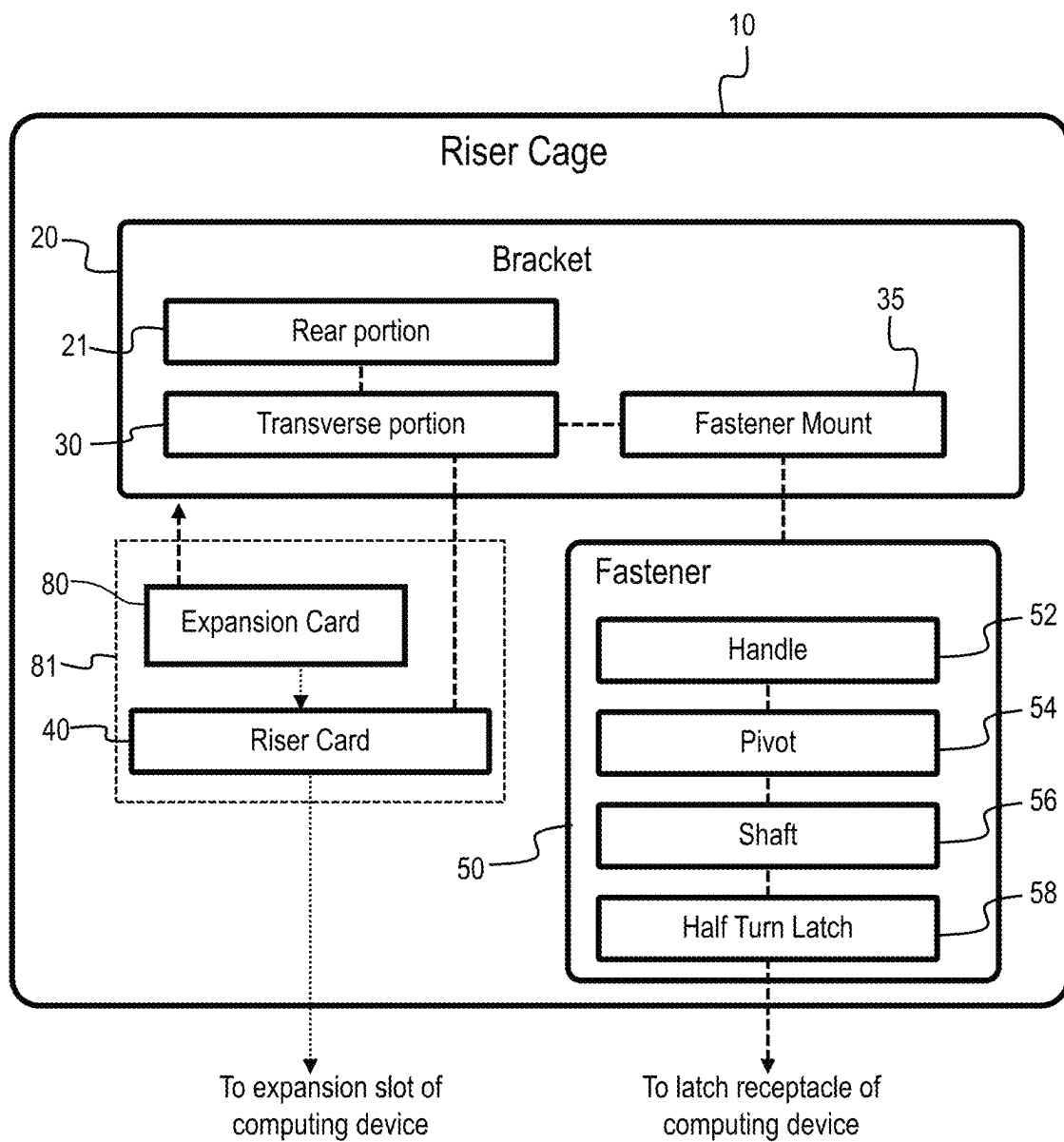
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FIG. 1



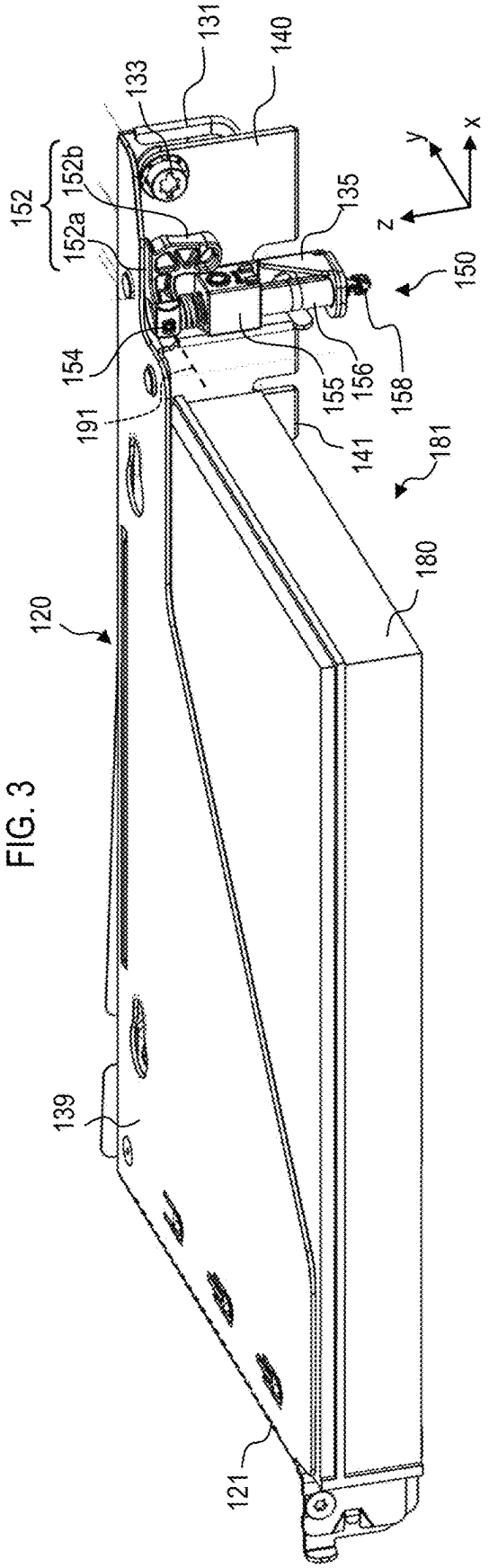
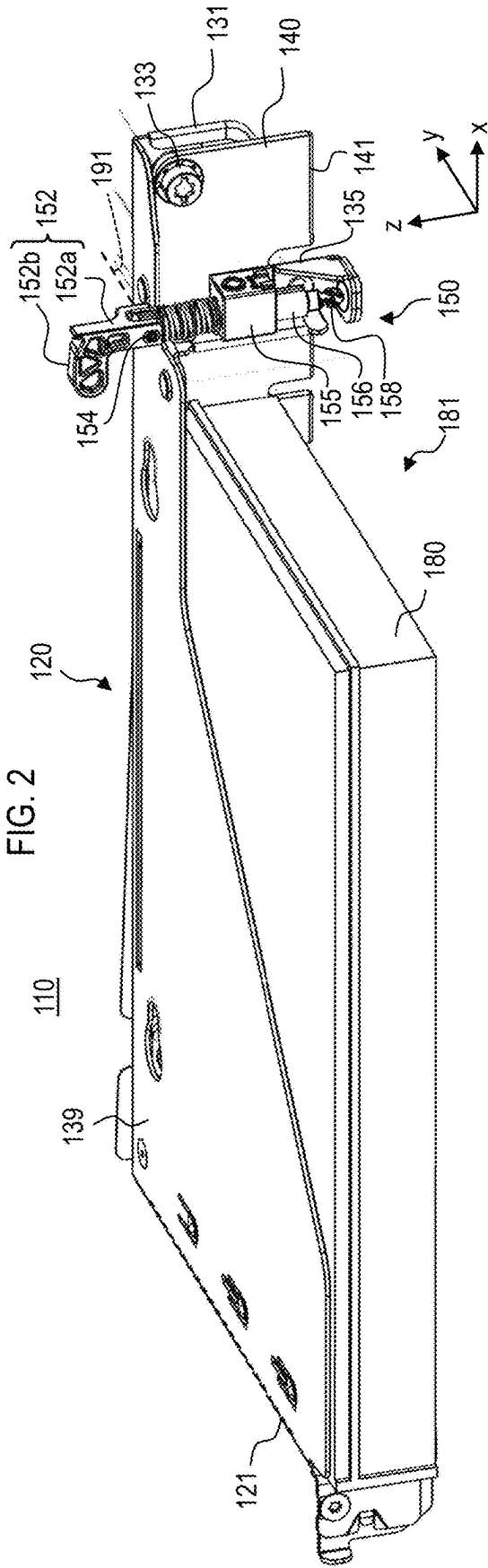


FIG. 4

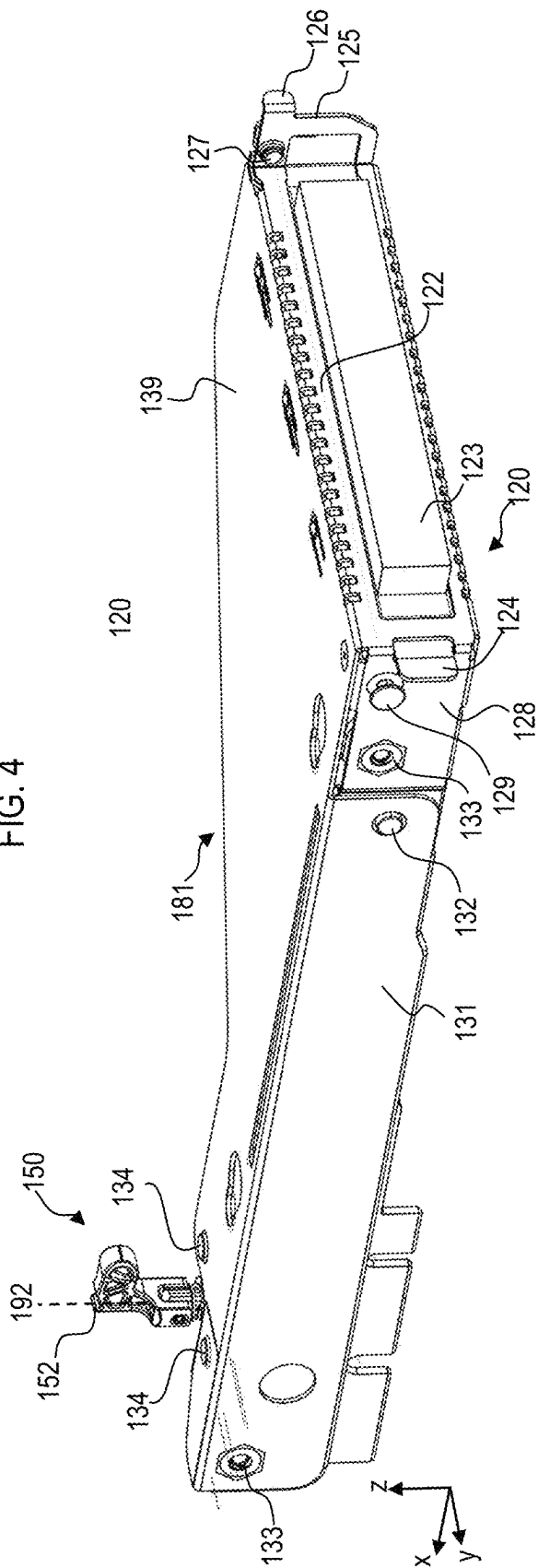
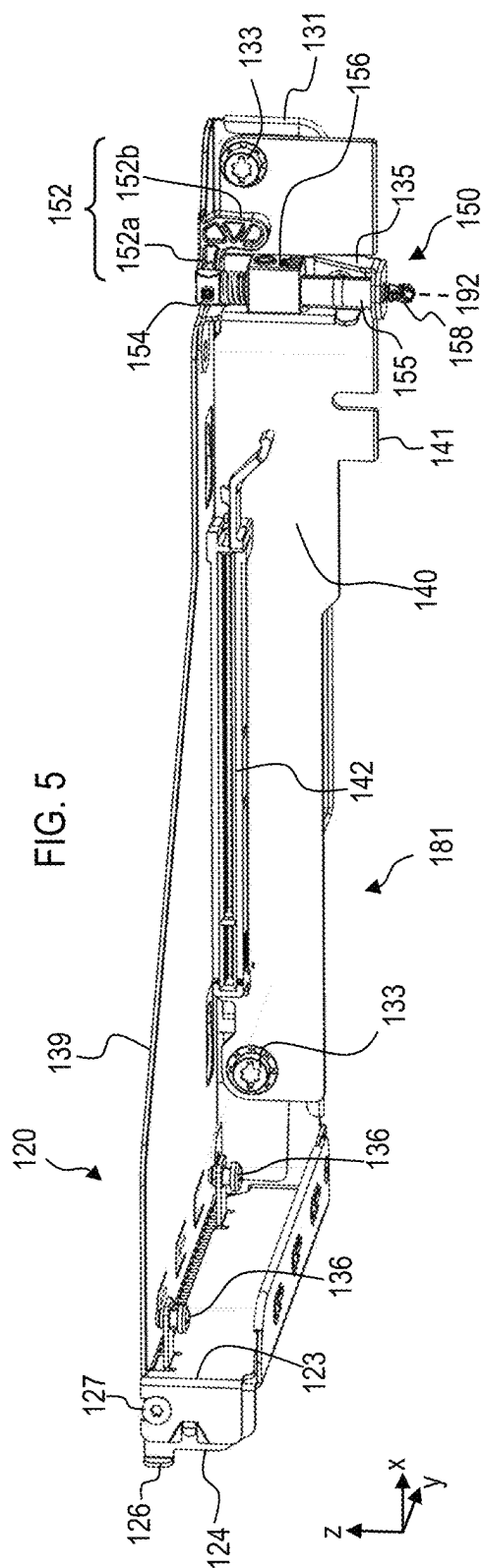


FIG. 5



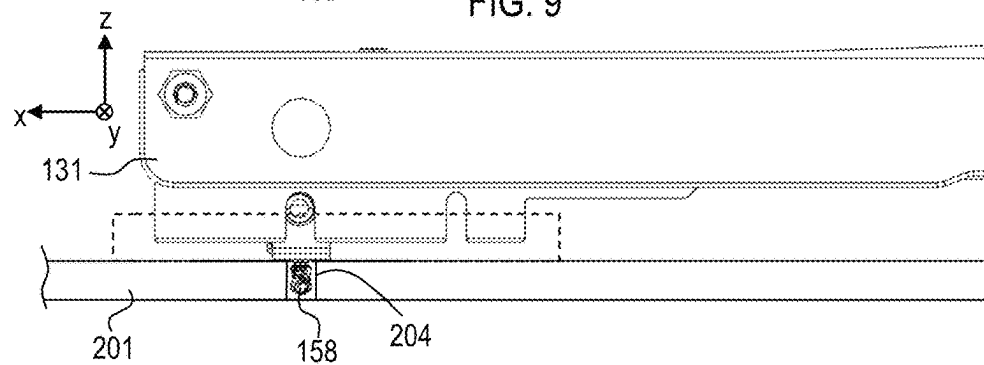
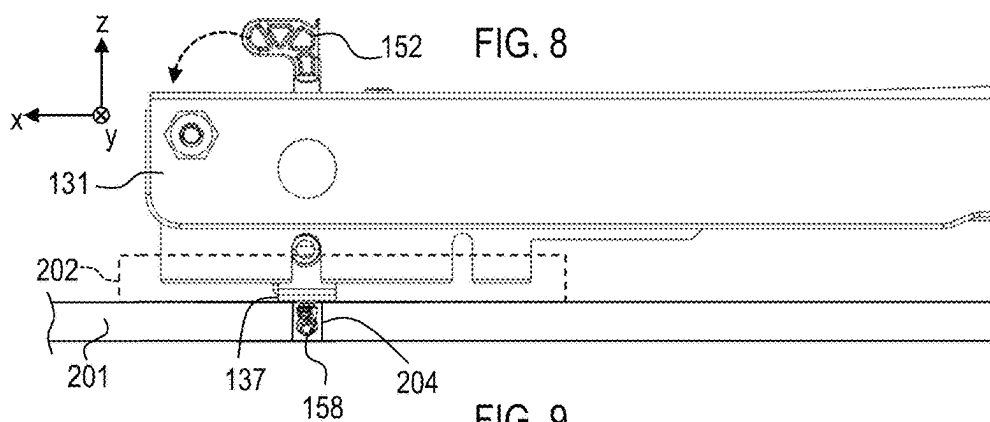
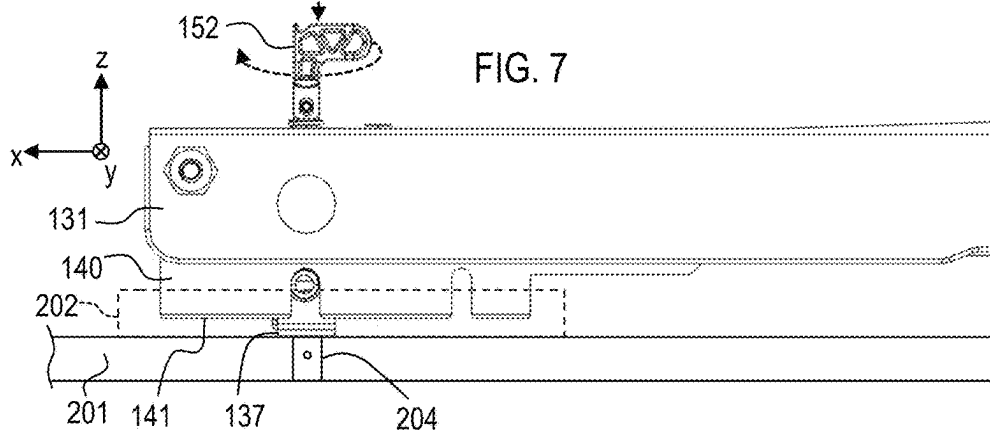
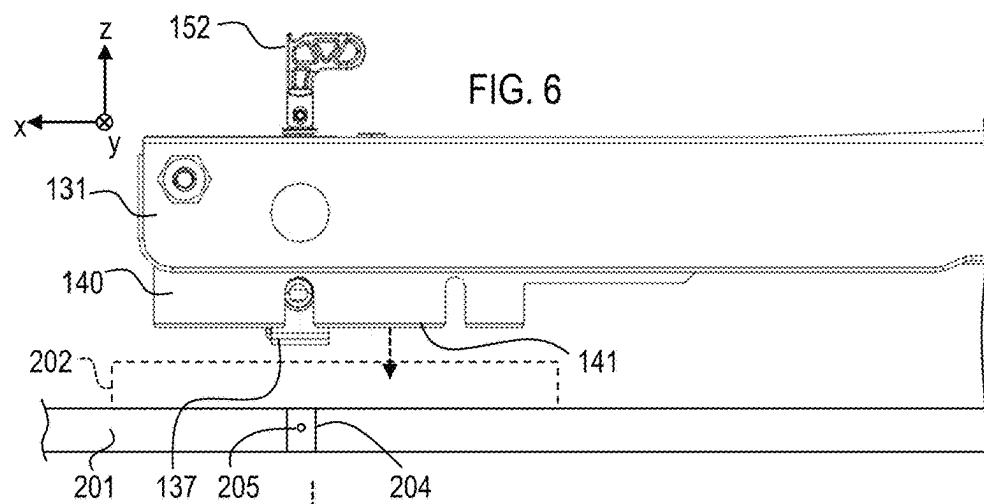


FIG. 10

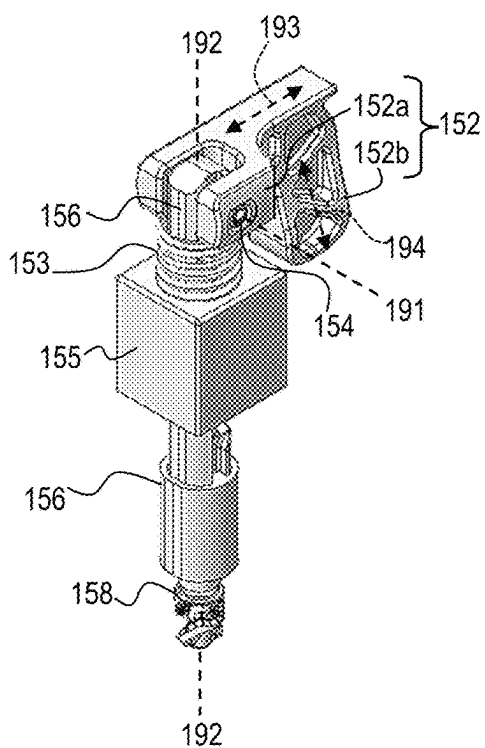


FIG. 11

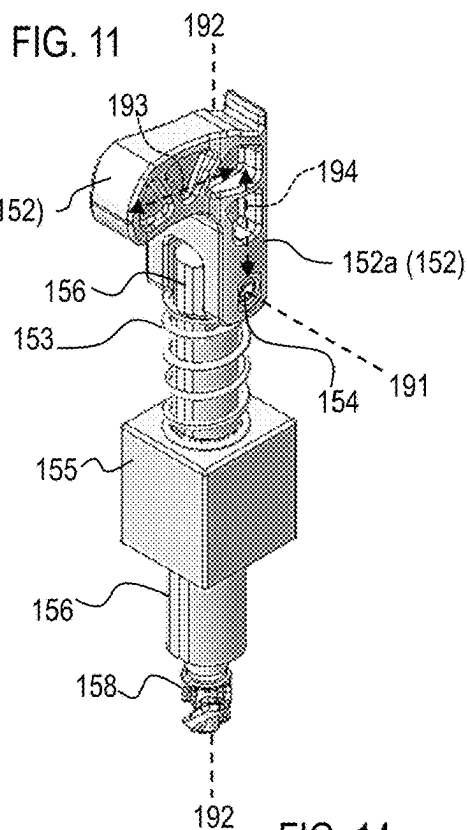


FIG. 12

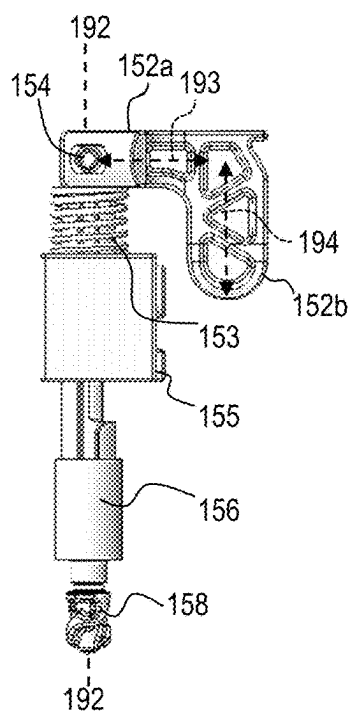


FIG. 13

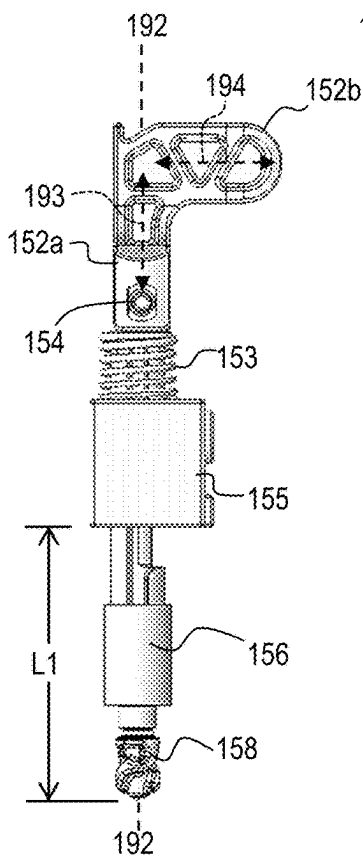
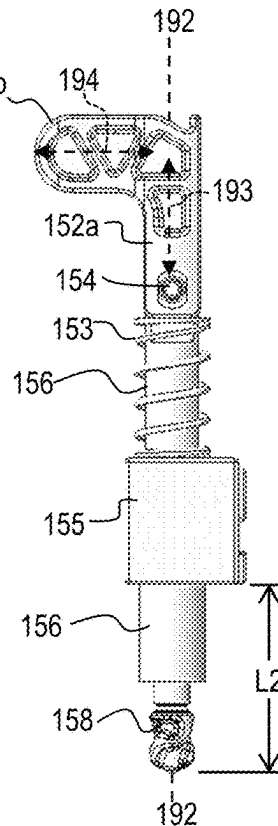


FIG. 14



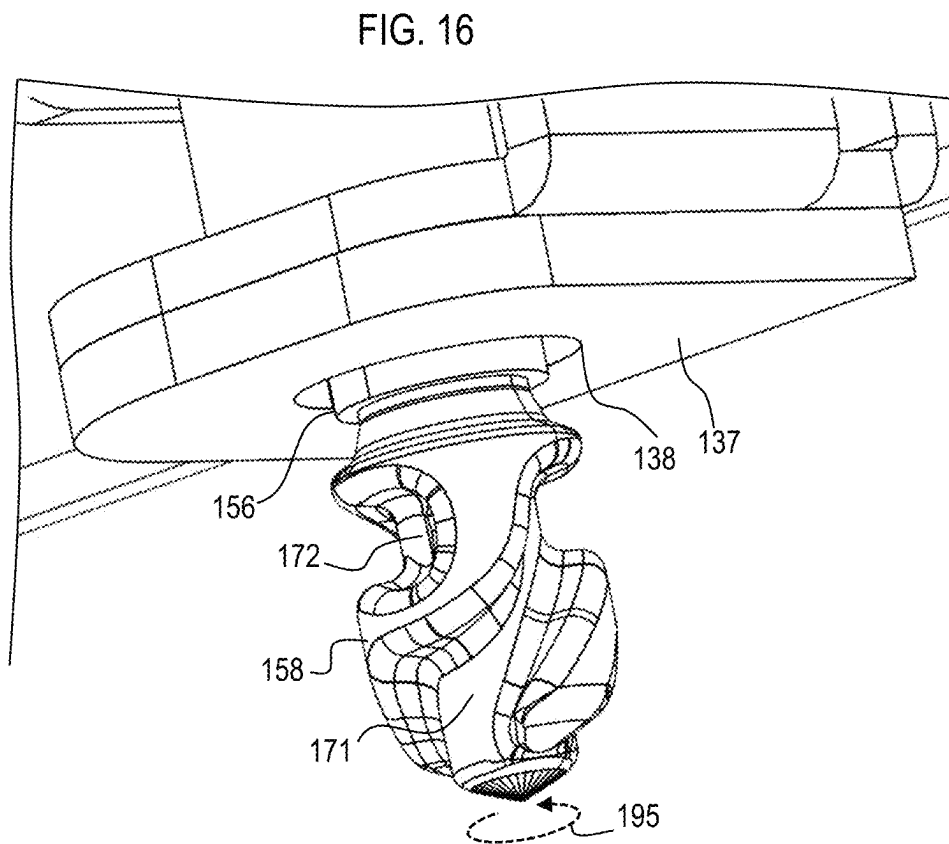
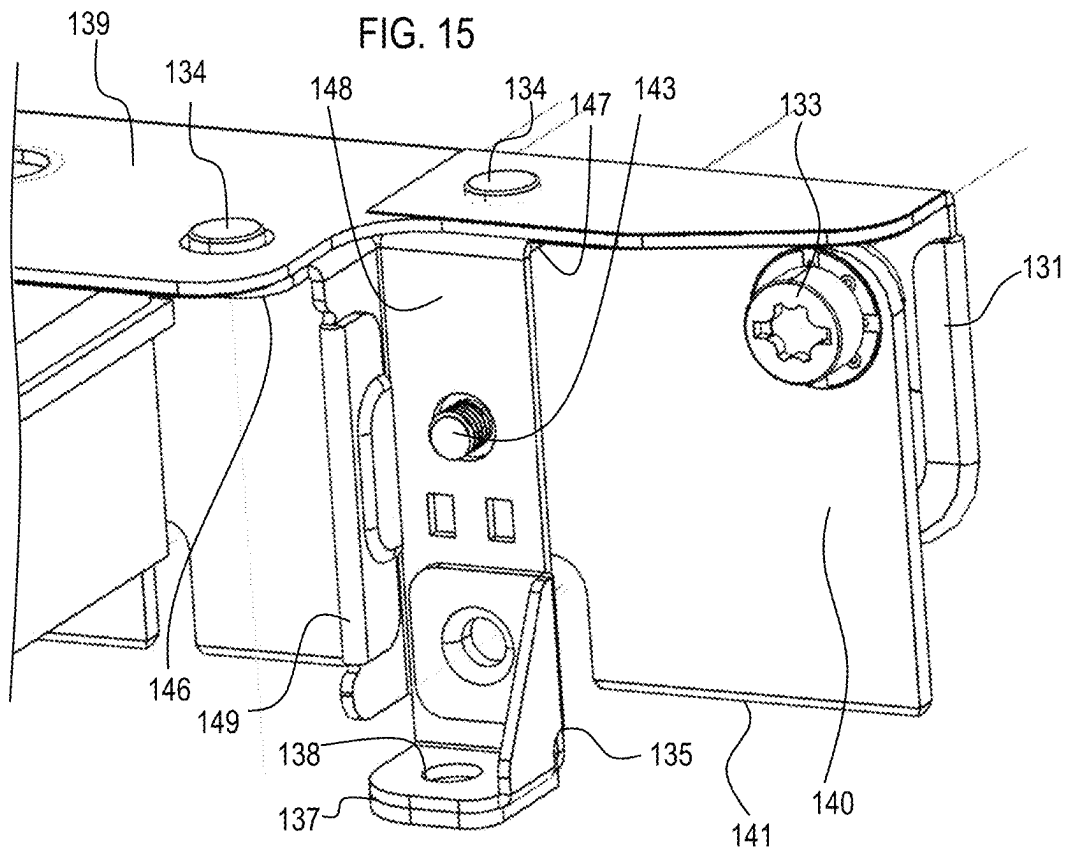
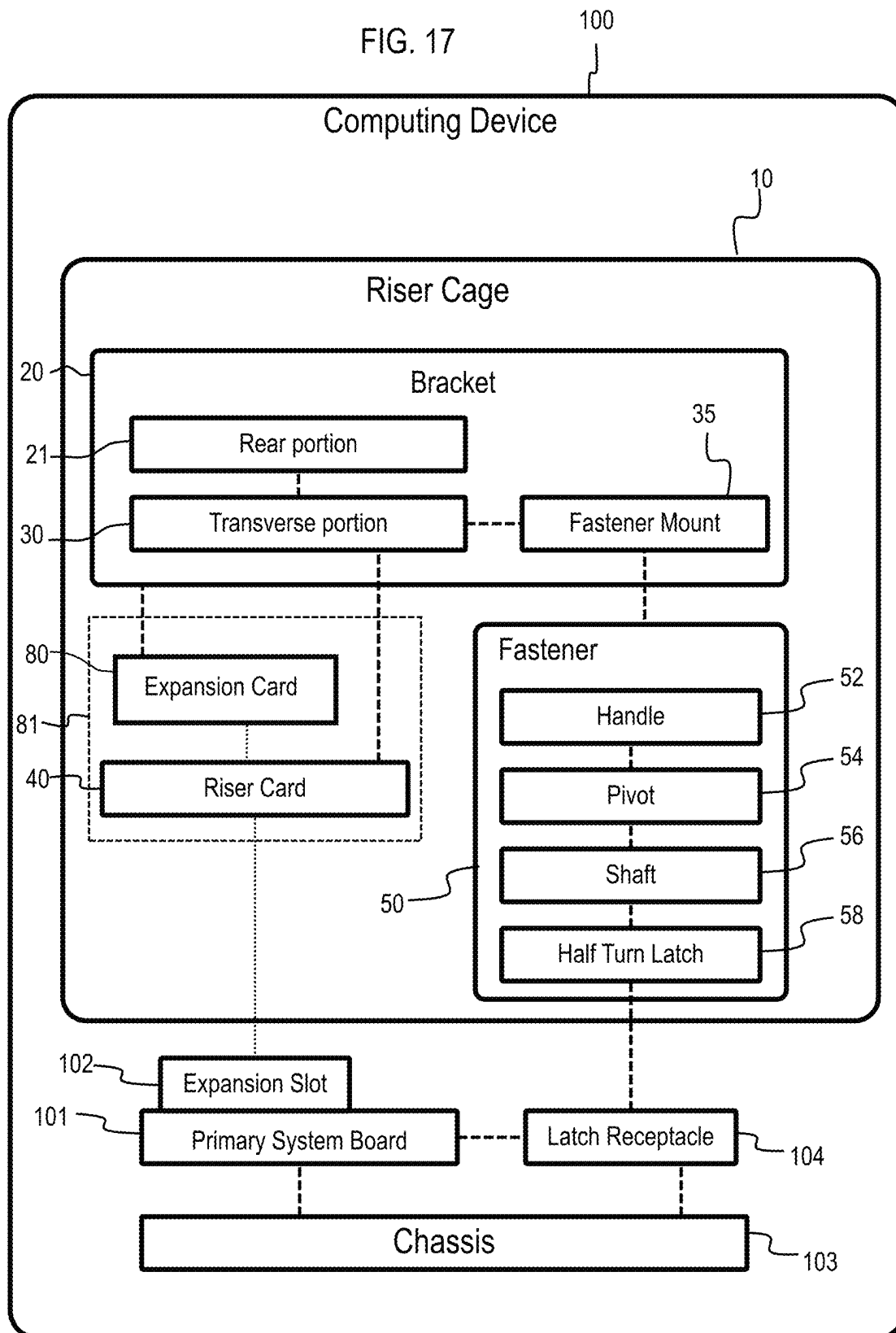


FIG. 17



RISER CAGE HALF TURN FASTENER**INTRODUCTION**

Computing devices, such as servers, generally have a primary system board (e.g., motherboard) which comprises a variety of components, such as power conversion circuitry, one or more CPU sockets to receive a processor, one or more memory sockets to receive memory devices, electrical and communications interconnects and busses, various controllers or chipsets (such as a baseboard management controller (BMC)), and others. In addition, to allow for more components to be added to the device, the primary system board may also include electrical connectors, which are configured to removably receive an electronic module. The addition of such modules generally expands the capability or functionality of the computing device, and thus the connectors and the modules have come to be known as expansion slots and expansion cards, respectively. Examples of such expansion slots include PCIe connectors, which come in different sizes and configurations (e.g., x16, x8, x1, etc.), M.2 connectors, and so on. Examples of common types of expansion cards include video cards or graphic processing units (GPUs), networking interface cards (NICs), storage controllers, hardware accelerators, and so on.

However, in some computing devices, it may not be possible to directly connect the expansion card to the expansion slot (e.g., due to orientation or space constraints), and thus the expansion card may be coupled to an intermediate board called a riser card, which is in turn connected to the expansion slot of the primary system board. The riser card is a printed circuit board assembly (PCA) which carries one electrical connector that is suitable for receiving the expansion card and another electrical connector suitable for being detachably connected to the expansion slot of the primary system board, and these connectors are arranged such that, when everything is connected, the expansion card fits as desired within the space constraints of the system.

For example, in some devices the expansion slots of the primary system board are oriented to receive expansion cards extending perpendicularly to the primary system board (a “vertical orientation”), but some computing devices (e.g., 1U or 2U servers) there may not be sufficient vertical clearance for an expansion card to have such a vertical orientation. However, the riser card may be much shorter than the expansion card, and thus the riser card can be connected to the expansion slot in the vertical orientation, and then the expansion card may be connected to the riser card in an orientation parallel to the primary system board (a “horizontal orientation”), thus allowing the expansion card to fit within the computing device. However, the electrical connections of the expansion slot and the riser card are generally not robust enough to physically support and secure the riser card and expansion card. Thus, riser cards and expansion cards are generally deployed with a supporting structure to support and secure them to the chassis and/or system board. This supporting structure is referred to as a riser cage.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure can be understood from the following detailed description, either alone or together with the accompanying drawings. The drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate one or more examples of the present

teachings and together with the description explain certain principles and operation. In the drawings:

FIG. 1 is a block diagram illustrating an example riser cage assembly.

FIG. 2 is a perspective view of an example riser cage assembly with a half-turn fastener in an unfastened state.

FIG. 3 is another perspective view of the example riser cage assembly of FIG. 2 with the half-turn fastener in a fastened state.

FIG. 4 is another perspective view of the example riser cage assembly of FIG. 2 with the half-turn fastener in the unfastened state and with expansion card omitted.

FIG. 5 is another perspective view of the example riser cage assembly of FIG. 2 with the half-turn fastener in the fastened state and with expansion card omitted.

FIG. 6 is a side view of the example riser cage assembly of FIG. 2 in a first state during an installation process.

FIG. 7 is a side view of the example riser cage assembly of FIG. 2 in a second state during an installation process.

FIG. 8 is a side view of the example riser cage assembly of FIG. 2 in a third state during an installation process.

FIG. 9 is a side view of the example riser cage assembly of FIG. 2 in a fourth state during an installation process.

FIG. 10 is a perspective view of the half-turn fastener of the riser cage assembly of FIG. 2 in isolation, with the half-turn fastener in the fastened state.

FIG. 11 is a perspective view of the half-turn fastener of the riser cage assembly of FIG. 2 in isolation, with the half-turn fastener in the unfastened state.

FIG. 12 is a side view of the half-turn fastener of the riser cage assembly of FIG. 2 in isolation, with the half-turn fastener in the fastened state with handle stowed.

FIG. 13 is a side view of the half-turn fastener of the riser cage assembly of FIG. 2 in isolation, with the half-turn fastener in a fastened state with handle deployed.

FIG. 14 is a side view of the half-turn fastener of the riser cage assembly of FIG. 2 in isolation, with the half-turn fastener in the unfastened state.

FIG. 15 is a perspective view of a portion of the riser cage assembly of FIG. 2 showing a fastener mount.

FIG. 16 is a perspective view of a portion of the riser cage assembly of FIG. 2 showing a half-turn latch of the half-turn fastener.

FIG. 17 is block diagram illustrating an example computing device with the riser cage assembly of FIG. 1 installed therein.

DETAILED DESCRIPTION

Riser cages are attached to the chassis and/or system board of the computing device via one or more attachment features, which generally include at least one actuatable fastener. In some circumstances, it may be desired for these fasteners to be manually actuatable without the use of tools, as this may make installation and removal of the riser cage easier. However, many existing toolless fasteners tend to require some free space around them to allow a user to grasp and actuate the fastener. However, in some space constrained computing devices, there may not be sufficient free space to accommodate certain toolless fasteners. While the amount of space that is needed for actuating such a fastener may seem small, computing devices tend to be very space-constrained and allocating even a small amount of space to the riser cage fastener may mean that other components of the system need to be left out or made smaller, which may not be feasible. Thus, it can be challenging to provide a riser

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cage fastener which is easy to install or remove but which can also fit within tight space constraints.

A technical solution to the aforementioned problems may include a half-turn fastener for a riser cage which has a handle which can be selectively stowed when not in use or deployed when it is needed to actuate the fastener. The half-turn fastener may comprise a half-turn latch, which engages with a receptacle anchored to the computing device to selectively latch to or unlatch from the receptacle. The fastener can be switched between a fastened state and an unfastened state by rotation of the fastener (and hence the half-turn latch) 180 degrees (a half turn). The handle is coupled to the half-turn latch via a shaft so that a user may fasten or unfasten the fastener by grasping the handle and turning the handle 180 degrees about an axis of the shaft. The handle is coupled to the shaft by a pivot pin, which allows the handle to also be pivoted about a pivot axis perpendicular to the shaft. This pivoting about the pivot axis allows the handle to be moved between the stowed and deployed configurations. In the deployed configuration the handle protrudes above the riser cage, whereas in the stowed configuration, the handle does not protrude above the riser cage. Thus, with the handle deployed above the riser cage, there may be sufficient space around the handle for the user to grasp and actuate the handle. Moreover, with the handle stowed, the space above the riser cage may be free of obstructions, and thus other items (such as a cover of the chassis or another riser cage) may be placed in this space. In this manner, the half-turn fastener disclosed herein may be easily actuable while also fitting within a tightly space constrained system.

Furthermore, in some examples, the handle is configured such that, in the stowed configuration, the handle is parallel to the riser card of the riser cage. For example, in some implementations the handle may have a proximal portion which extends from the shaft in a first direction and a distal portion which extends from the proximal portion in a second direction transverse to the first direction, thus giving the handle a generally "L" shaped profile, and the handle is configured such that in the stowed configuration this L-shape is parallel to the riser card (e.g., the plane defined by the first and second directions is parallel to the riser card). In some examples, in the stowed configuration, the proximal portion extends horizontally while the distal portion extends vertically downward, and conversely in the deployed configuration the proximal portion extends horizontally while the distal portion extends vertically upward. In particular, in some examples, in the stowed configuration, the proximal portion extends horizontally forward (i.e., away from a rear portion of the riser cage bracket). The aforementioned configurations of the handle keep the stowed handle from protruding laterally into the riser cage volume, thus allowing other components to be positioned in this space—for example, some expansion cards may extend into the space laterally adjacent to the fastener. By avoiding the stowed handle obstructing the other components, more components may be fit within a given space and/or the riser cage and the computing device as a whole to be made more compact. Moreover, as noted, in some examples the stowed handle may protrude in a forward direction, which in some circumstance may be a more desirable direction for the fastener to extend because in some computing devices the space forward of the fastener is usually not occupied and thus the stowed handle does not obstruct anything by protruding into this space. In addition, because there is usually some open space forward of the fastener, a user can easily extend their finger down through this open space to reach under the

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handle to move it to the deployed configuration. Moreover, in the stowed configuration, the pivot axis is perpendicular to the riser card so that while pivoting the handle from the stowed state to the deployed state, the handle remains parallel to the riser card throughout the motion, which avoids any interference with objects located laterally adjacent to the fastener (such as a large expansion card).

In contrast, in a riser cage with a fastener that has a stowable handle which is parallel to the system board and extends laterally when stowed, it may be difficult to achieve the same compactness described above. In such a system, the stowed handle may obstruct objects from being positioned laterally adjacent to the fastener, which may prevent larger components (e.g., expansion cards) from being used or may require the riser cage to be made longer to accommodate those components. Moreover, even in cases in which an expansion card is not directly obstructed by the handle, the expansion card may be close enough to the fastener that it is difficult for the user to reaching their finger under the handle to grasp the handle and move it to the deployed position.

Turning now to the figures, various devices, systems, and methods in accordance with aspects of the present disclosure will be described.

FIG. 1 is a block diagram conceptually illustrating a riser cage 10. It should be understood that FIG. 1 is not intended to illustrate specific shapes, dimensions, or other structural details accurately or to scale, and that implementations of the riser cage 10 may have different numbers and arrangements of the illustrated components and may also include other parts that are not illustrated. In FIG. 1, physical connections (attachment) are indicated conceptually by dashed lines, whereas electrical connections are indicated conceptually by dotted lines.

As shown in FIG. 1, the riser cage 10 comprises a bracket 20, a riser card 40 connected to the bracket 20, and a fastener 50 connected to the bracket 20. These components will be described in greater detail in turn below.

The bracket 20 may comprise a rear portion 21, a transverse portion 30, and a fastener mount 35. The rear portion 21 is configured to attach to a chassis of a computing device when the riser cage 10 is installed in the computing device. The transverse portion 30 is connected to the rear portion 21 and comprises one or more support structures oriented transverse to the rear portion 21. The rear portion 21 and the transverse portion 30 define a riser cage volume 81 in which an expansion card 80 may be received. The fastener mount 35 and riser card 40 are both attached to the transverse portion 30.

As noted above, the rear portion 21 attaches to the chassis of the computing device. More specifically, the rear portion 21 attaches either directly to the chassis or to other components which are, directly or indirectly, attached to the chassis. The rear portion 21 may form part of the chassis once attached thereto. For example, the rear portion 21 may form part of a rear panel of a chassis when installed. The rear portion 21 is described as the "rear" portion because in a typical installation scenario the riser cage 10 may be installed at a rear of the computing device with the rear portion 21 oriented towards a rearward direction, but this terminology is used merely for convenience, and does not limit the orientation of the riser cage 10 or the computing device in practice. For example, the riser card 10 could be installed at a front of a computing device with the rear portion 21 facing in a forward direction and forming a part of the front panel of the chassis. To enable attachment of the rear portion 21 to the chassis (or adjacent components) the

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rear portion **21** may comprise attachment features, such as tabs, spools, apertures, slots, or the like which engage with complementary attachment features of the chassis and/or of adjacent components to (directly or indirectly) secure the rear portion **21** to the chassis. The rear portion **21** may have a window panel which has perforations, slots, or other openings to allow airflow through the rear portion **21**. The rear portion **21** (specifically the window panel thereof) is generally parallel to a first plane, which is parallel to the front and rear panels of the chassis when the riser cage **10** is installed therein.

The transverse portion **30** is attached to the rear portion **21**. The transverse portion **30** comprises a side portion, a top portion, or both a side portion and a top portion. The side portion (when included) comprises a panel which is generally parallel to a second plane and is connected to the rear portion **21**. The second plane is perpendicular to the first plane and is also parallel to side walls of the chassis when the riser cage **10** is installed thereon. The top portion (when included) comprises a panel which is generally parallel to a third plane and connected to the rear portion **21**. The third plane is perpendicular to both the first and second planes and is also parallel to top and bottom walls of the chassis when the riser cage **10** is installed therein. In some examples in which both side and top portions are present, the top and side portions may be connected to one another and both may be connected to the rear portion **21**. In some examples, the transverse portion **30** is integrally connected to the rear portion **21**, or in other words the transverse portion **30** and the rear portion **21** are part of a unitary body (e.g., a single piece of sheet metal which has been bent, stamped, or otherwise formed into the shape of the bracket **20**).

As noted above, the fastener mount **35** is coupled to the transverse portion **30**. The fastener mount **35** may be positioned near a front end of the bracket **20**, which is the end opposite from the rear portion **21**.

The riser card **40** is disposed within the riser cage volume **81** and is physically attached to the transverse portion **30** of the bracket **20**. In some examples in which the transverse portion **30** comprises a side portion, the riser card **40** is attached to the side portion. The riser card **40** comprises a PCA with two electrical connectors: a first connector configured to electrically connect to an expansion slot of the computing device and a second connector configured to electrically connect to an expansion card **80** disposed within the riser cage volume **81** of the riser cage. The riser card **40** is oriented parallel to the transverse portion **30** (parallel to the second plane).

The fastener **50** is connected to the fastener mount **35**. The fastener **50** is configured to engage with a latch receptacle of the computing device to secure the riser cage to the chassis (directly or indirectly). The latch receptacle may be coupled to the system board, a support structure which supports the system board, or to the chassis. In some examples, the fastener **50** is not the sole mechanism which supports and secures the riser cage **10** to the chassis. As noted above, in some examples the rear portion **21** comprises attachment features to connect to the chassis. Thus, in such examples, the attachment features at the rear portion **21** may support and secure the rear of the riser cage **10** to the chassis, whereas the fastener **50** may support and secure the front side of the riser cage **10**. In addition, the other attachment features, when present, may not fully secure the riser cage **10** to the chassis in all cases. For example, the attachment features may be configured such that they constrain the riser cage **10** in some directions but allow movement of the riser cage **10** in certain other directions, and such movement of

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the riser cage **10** may result in the attachment features disengaging from the chassis. Accordingly, one purpose of the fastener **50** in such examples is to prevent such movements of the riser cage **10** that would otherwise cause the other attachment features to disengage. In this manner, the attachment features and the fastener **50** may work together to secure the riser cage **10** to the chassis.

The fastener **50** comprises a half-turn latch **58** at one end and a handle **52** at the opposite end, with a shaft **56** connecting the two. The half-turn latch **58** is configured to be extended into a latch receptacle of the computing device when the riser cage **10** is installed therein and engage with latch elements in the receptacle to selectively latch to or unlatch from the receptacle. The fastener **50** is configured to be changeable between a fastened state and an unfastened state by rotation and translation of the fastener **50** about and along the axis of the shaft **56**.

When the riser cage **10** is positioned in, but not yet attached to, the computing device, transitioning the fastener to the fastened state comprises extending the half-turn latch **58** into the receptacle (e.g., by causing the fastener **50** to move downward along the axis of the shaft **56**) and then rotating the half-turn latch **58** within the receptacle by rotating the fastener **50** in a first rotational direction until it reaches a first orientation. The half-turn latch **58** may comprise engagement features (e.g., grooves) which progressively engage with complementary engagement features (e.g., protrusions) of the latch receptacle as the fastener **50** is rotated. A fully latched configuration between these engagement features is achieved when the fastener **50** reaches the first orientation, whereupon the half-turn latch **58** (and hence the fastener **50** as a whole) is latched to the receptacle. This in turn secures the riser cage **10** to the computing device.

Conversely, the fastener **50** can be transitioned from the fastened state to the unfastened state by rotating the fastener **50** in a second rotational direction, opposite the first rotational direction, until the fastener **50** reaches a second orientation. As the fastener **50** (and hence half-turn latch **58**) is rotated in the second rotational direction, the engagement features and complementary engagement features move out of the latched configuration and progressively become disengaged until eventually being fully unfastened when the fastener **50** reaches the second orientation. At this point, the half-turn latch **58** is unlocked from the receptacle and can be freely removed therefrom. The first orientation and the second orientation may be 180 degrees apart, so that turning the fastener **50** a half turn (180 degrees) changes the fastener between the first orientation (fastened state) and the second orientation (unfastened state).

As noted above, the fastener **50** comprises a handle **52**, which is pivotably coupled to the shaft **56** via pivot **54**. The handle **52** is graspable by a user to facilitate the user turning the fastener **50** about the axis of the shaft **56**. In particular, the handle **52** may have a proximal portion which extends along a first direction and a distal portion which extends from the proximal portion along a second direction transverse to the first direction, thus giving the handle a generally "L" shaped profile. The first direction and the second direction (which correspond to the directions of extend of the proximal and distal portions of the handle **52**) together define a plane, which may be referred to herein as the handle plane. The distal portion of the handle provides leverage to the user, making it easier for them to fasten and unfasten the fastener **50**.

The handle **52** is coupled to the shaft by a pivot **54**, which allows the handle to be pivoted about a pivot axis perpen-

dicular to the axis of the shaft **56**. This pivoting about the pivot axis allows the handle **52** to be moved between stowed and deployed configurations. In the deployed configuration the handle **52** protrudes above the riser cage **10**, whereas in the stowed configuration, the handle **52** does not protrude above the riser cage **10**. Generally, it is unlikely for there to be an obstruction in the region directly above the riser cage **10** when the handle is being deployed, because any components that would normally occupy this region (such as a cover of the computing device or another riser cage) are removed from the device prior to attempting to install or remove the riser cage **10**. Thus, in the deployed state there may be free space all around the handle **52** which allows the user to easily grasp and rotate the handle **52**. Moreover, once the riser cage **10** is installed and secured and it is desired to return the computing device to its normal state, the handle **52** can be stowed and thereafter it ceases to protrude above the riser cage **10**, thus allowing the other components which would normally occupy that region to be returned to their normal place. In this manner, the half-turn fastener **50** may be easily actuatable when deployed but then can be stowed when not needed to make room for other components, thus allowing for a relatively more compact (space constrained) system to be designed.

In addition, handle **52** is configured such that, in the stowed state, the handle **52** is parallel to the riser card **40** and/or side portion of the bracket **20**. References to the handle **52** being parallel to the riser card **40** and/or side portion side portion of the bracket **20** mean that the handle plane defined by the handle **52** (as defined above) is parallel to a riser card **40** or side portion, or in other words parallel to the second plane. This configuration of the handle **52** allows the stowed handle **52** to not protrude laterally into the riser cage volume. Instead, the handle **52** may protrude in a forward direction (away from the rear portion **21** of the riser cage **10**). This protrusion in a forward direction is generally more desirable than protrusion in a lateral direction, as the region directly forward of the fastener **50** is usually not occupied, whereas it may be desired to occupy the region directly laterally adjacent the fastener **50** with other objects, such as the expansion card. Moreover, in the stowed state, the pivot axis is perpendicular to the riser card **40** so that when pivoting the handle **52** from the stowed state to the deployed state the handle **52** remains parallel to riser card **40** throughout the motion. Thus an object (e.g., expansion card) can be disposed laterally adjacent to the fastener **5** without interfering with the pivoting of the handle **52** between deployed and stowed configurations. In addition, because there is usually open space forward of the fastener **50**, a user can easily extend their finger down through this open space to reach under the handle **52** to grasp and deploy it.

In some examples, the expansion card **80** is included in the riser cage **10** (within the riser cage volume **81**), with the expansion card **80** being electrically connected to the riser card **40** (as indicated by the dot-lined arrow in FIG. **1**) and physically attached to the bracket **20** (as indicated by the dash-lined arrow in FIG. **1**). In other examples, the riser cage **10** does not include the expansion card **80**; in other words, although the riser cage **10** may be intended for use with the expansion card **80**, in some cases the riser cage **10** may be provided (e.g., manufactured, sold, installed in a computing device, etc.) separately from the expansion card **80**.

As shown in FIG. **17**, in some examples, a computing device **100** comprises the riser cage **10** installed therein. The computing device **100** comprises a chassis **103**, a primary system board **101** attached (directly or indirectly) to and supported by the chassis **103**, an expansion slot **102** mounted

to the primary system board **101**, and a latch receptacle **104** attached to the chassis **103** and/or to primary system board **101**. In the installed state of the riser cage **10** in the computing device **100**, the riser card **40** is electrically connected to the expansion slot **102**, and the fastener **50** is engaged with and fastened to the latch receptacle **104**. The expansion card **80** is disposed in the riser cage volume **81**, is electrically connected to the riser card **40**, and is attached to the bracket **20**.

Turning now to FIGS. **2-16**, another example riser cage **110** will be described. The riser cage **10** is one example configuration of the riser cage **10**. Thus, some components of the riser cage **110** correspond to (i.e., are the same as, similar to, and/or example configurations of) components of the riser cage **10** described above. The components of the riser cage **110** and the components of the riser cage **10** which correspond to one another are given reference numbers with the same last two digits, such as **50** and **150**. The descriptions above of the components of the riser cage **10** are applicable to the corresponding components of the **110** (unless otherwise indicated or logically contradictory), and thus duplicative descriptions of some aspects already described above are omitted below. In some examples, the riser cage **10** included in the computing device **100** of FIG. **17** may comprise the riser cage **110**.

FIGS. **2-16** show the riser cage **110** in various perspectives and in various states, and aspects of the riser cage **110** may be visible in multiple of the figures. The description below will refer to the figures as and when they are relevant to the aspect being described, rather than in strict numerical order. Various aspects of the riser cage **110** are described herein in relation to the X, Y, and Z axes illustrated in FIGS. **2-16**. In addition, herein an XY plane may be referred to, which is a plane defined by the X and Y axes. Similarly, an XZ plane may be referred to, which is a plane defined by the X and Z axes. Also, a YZ plane may be referred to, which is a plane defined by the Y and Z axes. Things which are parallel to the XY plane may be described herein as "horizontal," whereas things which are parallel to either the XZ or YZ plane may be described as "vertical." Occasionally, a structure is referred to herein as being parallel or perpendicular to a plane (or to another structure), and it should be understood that such references refer to the orientation of a broadest face of the structure (i.e., not to edges).

As shown in FIGS. **2-5**, the riser cage **110** comprises a bracket **120**, a riser card **140** connected to the bracket **120**, and a fastener **150** connected to the bracket **120**.

The bracket **120** comprises a rear portion **121**, a side portion **131**, a top portion **139**, and a fastener mount **135**. The rear portion **121** is configured to attach to a chassis of a computing device when the riser cage **110** is installed in the computing device. The side portion **131** and top portion **139** are both connected to the rear portion **121** and to one another (the side portion **131** and top portion **139** together are an implementation example of the transverse portion **30** in FIG. **1**). The rear portion **121**, side portion **121**, and top portion **139** define a riser cage volume **181** in which an expansion card **180** may be received. The fastener mount **135** is attached to the top portion **139**, and the riser card **140** is attached to the side portion **131**. These components will be described in greater detail in turn below.

FIGS. **4** and **5** illustrate the rear portion **121** of the bracket **120**. The rear portion **121** comprises a window panel **122**. The window panel **122** is generally parallel to a first plane. This first plane is parallel to the YZ plane. The first plane is also parallel to the front and rear panels of the chassis when the riser cage **110** is installed therein. The window panel **122**

is configured to receive a window insert **123**. The window insert **123** is coupled to the window panel **122** via attachment features **124** and **127**. In the illustrated example, the attachment feature **124** comprises a tab that is inserted through a slot in the window panel **122**, and the attachment feature **127** is a screw that engages an aperture in a flange **125** attached to the window panel **122**. In other examples, other attachment features may be used. The window insert **123** may have perforations, slots, or other openings (not illustrated) to allow airflow through the rear portion **121**. As shown in FIG. 5, the window insert **123** comprises attachment features **136** which are configured to attach to the expansion card **180**. Thus, the expansion card **180** is secured to the window insert **123**, which is in turn secure to the window panel **122**.

The rear portion **121** also comprises attachment features **126** and **129**, which are best seen in FIG. 4. In the illustrated example, the attachment features **126** and **129** comprise a tab and a spool, respectively, which are configured to engage with complementary attachment features of the chassis or of an adjacent component of the computing device. In other examples, any other type of attachment features may be used.

As shown in FIG. 4, the rear portion **121** is connected to the side portion **131** and to the top portion **139**. In the illustrated example, the top portion **121** is integrally connected to the window panel **122** (e.g., they are formed from the same unitary body). In other examples, they may be separate parts which are connected together via mechanical fasteners, welding, adhesives, friction fittings, or any other fastening technique. The rear portion **121** also comprise a wing **128** which is attached to the window panel **122** but which extends perpendicular to the window panel **122**. For example, the wing **128** may be integrally connected to the window panel **122** and may be bent into the position shown in FIG. 4. The wing **128** may partially overlap with the side portion **131** (in FIG. 4, the end (not visible) of the wing **128** is tucked behind the side portion **131**) and the overlapping portions of the wing **128** and side portion **141** may be attached via fastener **132**. In other examples, the wing **128** may be attached to the side portion **141** by welding, adhesives, interlocking engagement, or any other fastening technique.

As shown in FIGS. 2-5, the side portion **131** comprises a panel which is generally parallel to a second plane, which is perpendicular to the first plane. This second plane is parallel to the XZ plane, which is parallel to side walls of the chassis when the riser cage **110** is installed thereon. The top portion **139** comprises a panel which is generally parallel to a third plane, which is perpendicular to the first plane and to the second plane. The third plane is parallel to the XY plane, which is parallel to top and bottom walls of the chassis when the riser cage **110** is installed therein. In the illustrated example, the top portion **139** is integrally connected to the side portion **131**. Thus, in the illustrated example, the rear portion **121**, side portion **131**, and top portion **139** of the bracket **120** are all part of a unitary body (e.g., a single piece of sheet metal which has been bent, stamped, or otherwise formed into the shape of the bracket **120**) and are all mutually perpendicular to one another.

As shown in FIGS. 2-5, the riser card **140** is connected to the side portion **131** via fasteners **133**. The riser card **140** is parallel to the XZ plane and to the side portion **131**. As shown in FIG. 5, the riser card comprises a first electrical connector **141** and a second electrical connector **142**. The first connector **141** is configured to mate with and electrically connect to a complementary electrical connector on the

primary system board of the computing device when the riser cage **110** is installed therein. For example, FIGS. 6-9, which are described in greater detail below, illustrate schematically the first connector **141** mating with the expansion slot **202**, which is a complementary electrical connector connected to a primary system board **201**. In the illustrated example, the first connector **141** is an edge connector (also called a PCB edge connector or PCB gold finger connector), whereas the expansion slot **202** is a female receptacle which receives the first connector **141**. In some examples, the first connector **141** is a PCIe edge connector configured to mate with a PCIe socket. The second connector **142** is configured to mate with and electrically connect to a complementary electrical connector on the expansion card **180** when the expansion card **180** is installed in the riser cage **110**. As shown in FIG. 5, in the illustrated example, the second connector **142** is a female receptacle which is to receive an edge connector of the expansion card **180**. In some examples, the second connector **142** is a PCIe socket.

As shown in FIGS. 2, 3, and 10-14, the fastener **150** comprises a half-turn latch **158** at one end and a handle **152** at the opposite end, with a shaft **156** connecting the two. The fastener **150** also comprises a housing **155** through which the shaft **156** extends (via a bore). The housing **155** connects the fastener **150** to the fastener mount **135** while allowing the fastener **150** to rotate and translate relative to the fastener mount **135**, as will be described in greater detail below. The half-turn latch **158** is configured to be extended into a latch receptacle **204** of the computing device when the riser cage **110** is installed therein and engage with latch elements **205** in the receptacle **204** (see FIG. 8) to selectively latch to or unlatch from the receptacle **204**. The fastener **150** is configured to be changeable between a fastened state and an unfastened state by 180-degree rotation of the fastener **150** about the axis **192** of the shaft **156**, as will be described in greater detail below.

As mentioned above and as shown in FIGS. 2-4, the fastener **150** comprises a handle **152** coupled to the shaft **156**. The handle **152** is graspable by a user to facilitate the user turning the fastener **150** about the axis **192** (see FIG. 4). In particular, as best seen in FIGS. 10-14, the handle **152** may have a proximal portion **152a**, which extends along a first direction **193** and a distal portion **152b** which extends from the proximal portion **152a** along a second direction **194** transverse to the first direction **193**, thus giving the handle a generally "L" shaped profile. The first direction **193** and the second direction **194** together define a plane, which may be referred to herein as the handle plane. The handle plane may change orientations as the fastener is transitioned between states, as will be described below. The distal portion **152b** of the handle **152** provides leverage to the user, making it easier for them to rotate the fastener **150** to fasten and unfasten the fastener **150**.

The handle **152** is coupled to the shaft by a pivot **154** (e.g., pivot pin), which allows the handle **152** to be pivoted about a pivot axis **191** perpendicular to the axis **192** of the shaft **156**. This pivoting about the pivot axis **191** allows the handle **152** to be moved between a stowed configuration and a deployed configuration, which will be described below.

FIGS. 2, 4, 6-8, 11, 13, and 14 illustrate the handle **152** in the deployed configuration. As shown in FIG. 2, in the deployed configuration, the proximal portion **152a** of the handle **152** extends vertically upward (i.e., in the +Z direction). In addition, in the deployed configuration the distal portion **152b** extends in any horizontal orientation (i.e., any orientation parallel to the XY plane). As shown in FIG. 2, in the deployed configuration the handle **152** protrudes upward

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(in the +z direction) above the riser cage **110**. Generally, it is unlikely for there to be an obstruction in the region directly above the riser cage **110** when the handle **152** is in the deployed configuration because the components which would normally occupy this region (such as a cover, another riser cage, or other component) are removed prior to attempting to install or remove the riser cage **110**. Thus, with the handle **152** protruding above the riser cage **110** in the deployed state, there may be sufficient free space around the handle **152** for the user to easily grasp and rotate the handle **152**.

FIGS. 3, 5, 9, 10 and 12 illustrate the handle **152** in the stowed configuration. As shown in FIG. 3, in the stowed configuration, the proximal portion **152a** of the handle **152** extends horizontally forward (i.e., away from the rear portion **121**, or in the +x direction), while the distal portion **152b** extends vertically downward (i.e., -z direction). In addition, as shown in FIGS. 3, 5, and 9, in the stowed configuration, the handle **152** does not protrude significantly (in some examples, not at all) above the riser cage **110**. Thus, after the riser cage **110** has been installed and secured, the handle **152** can be stowed and thus the handle **152** ceases to occupy the space above the riser cage **110**, allowing for the other components which were previously removed to be returned to their normal positions. Thus, the stowed state allows the fastener **150** to be compacted during normal system operation so as to not obstruct the other items in the computing device, while the deployed state allows the fastener **150** to be uncompacted during installation/removal to facilitate easier actuation of the fastener **150**.

In addition, handle **152** is configured such that, in the stowed state, the handle **152** is parallel to the riser card **140** and side portion **131**. In other words, in the stowed position, the handle plane is parallel to the XZ plane. Thus, the stowed handle **152** does not protrude laterally in the -y direction into the riser cage volume **181**. Instead, the handle **152** may protrude in a forward (+x) direction, as described above. This protrusion in a forward direction is generally more desirable than protrusion in a lateral direction, as the region directly forward of the fastener **150** is usually not occupied, whereas it may be desired to occupy the region directly laterally adjacent the fastener **150**. For example, although a shorter expansion card **180** is illustrated in FIG. 3 to allow for better visibility of the fastener **150**, the riser cage **110** may also be capable of receiving longer expansion cards which extend further in the +x direction, and thus are positioned laterally adjacent to the fastener **150**. In addition, because there is usually open space forward of the fastener **150**, a user can easily extend their finger down through this open space to reach under the handle **152** to grasp and deploy it.

Moreover, in the stowed state, the pivot axis **191** is perpendicular to the riser card **140**, as shown in FIG. 3, and thus as the handle **152** pivots from the stowed state to the deployed state, the handle **152** remains parallel to the ZY plane throughout the motion. This allows the deployment of the handle **152** to not be obstructed by any objects which happen to be positioned laterally adjacent to the fastener **150**, such as the large expansion card described above.

As mentioned above, the fastener can be transitioned between fastened and unfastened states. FIGS. 2, 4, 6, 11, and 14 illustrate the fastener **150** in the unfastened state. The unfastened state is characterized by the fastener **150** being oriented at a first rotational orientation. More specifically, in the first rotational orientation, the distal portion **152b** of handle **152** points rearward towards the rear portion **121** (i.e., in a-x direction), as shown in FIGS. 2 and 6. In

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addition, in some examples, in the unfastened state, the fastener **150** may be at a retracted position. As shown in FIG. 14, in this retracted position, the tip of the latch **158** is located a distance **L2** from the bottom of the housing **155**, which is higher in the +z direction than a lowest possible position for the latch **158**. In some examples, the retracted position is one in which the latch **158** does not extend fully into the receptacle **204** and/or is retracted above a bottom of the fastener mount **135** (i.e., above the floor **137**), as shown in FIGS. 2 and 7. A spring **152** may be provided around the shaft **156**, between the housing **155** and the handle **152**, to bias the fastener **150** towards the retracted position.

FIGS. 3, 5, 8, 9, 10, and 12 illustrate the fastener **150** in the fastened state. The fastened state is characterized by the fastener **150** being oriented at a second rotational orientation and at an extended position. More specifically, in the second rotational orientation, the distal portion **152b** of handle **152** points forward (i.e., away from the rear portion **121**, in a +x direction), as shown in FIGS. 3 and 8. The extended position of the fastener **150** comprises a position in which the latch **158** is extended farther downward (in the -z direction) than in the retracted position. In particular, in some examples, the extended position is one in which the latch **158** is extended sufficiently far downward (in the -z direction) to be received into the receptacle **204** (assuming the riser cage **110** is in the installation position in the computing device) and/or extends beyond a bottom of the fastener mount **135** (e.g., beyond shelf **137**), as shown in FIGS. 3 and 8. FIG. 13 illustrates the fastener **150** in the extended position, with the tip of the latch **158** being located a distance **L1** from the bottom of the housing **155**. The distance **L1** in the extended position (FIG. 13) is greater than the distance **L2** in the retracted position (FIG. 14).

The transitioning between these states will be described in greater detail below with reference to FIGS. 6-14, which show a sequence of events in an installation of the riser cage **110** in a computing device comprising a primary system board **201**. FIG. 6 illustrates a first state, in which the riser cage **110** is brought to an installation position while the fastener **150** is in the unfastened state. In this state, the first connector **141** of the riser card **140** is aligned with the complementary connector **202** of the system board **201**, and a shelf **137** of the fastener mount **135** is also aligned with a receptacle **204** which is coupled to the system board **201** (the receptacle **204** could instead be coupled to a support structure or chassis).

The riser cage **110** may then be moved towards the system board **201** along the -z direction as indicated by the dashed arrow in FIG. 6, resulting in the state illustrated in FIG. 7. In this state illustrated in FIG. 7, the first electrical connector **141** is mated with the complementary connector **202**. In addition, the fastener **150** remains in the unfastened state. In some examples, the shelf **137** is adjacent to, and in some examples abutting, the system board **201** and/or the receptacle **204**. Although not illustrated, during the transition from the state of FIG. 6 to the state of FIG. 7, the other attachment features **126** and **129** of the riser cage may be engaged with the chassis or other adjacent components.

The fastener **150** may then be transitioned to the fastened state. This comprises moving the fastener **150** downward along axis **192** until the half-turn latch **158** is in the extended position and then rotating the fastener **150** (via the handle **152**) 180 degrees about the axis **192** of the shaft **136** until the handle **152** faces forward (+x direction), as indicated by the dashed arrows in FIG. 7. This results in the state illustrated in FIG. 8, in which the half-turn latch **158** has extended into the receptacle **204** and engaged with the latch elements **205**

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so as to secure the half-turn latch **158** to the receptacle **204**. In other words, the fastener **150** is in the fastened state.

More specifically, as shown in FIG. **16**, the half-turn latch **158** may comprises grooves **171** that spiral around and up the latch **158** (like a helix). During the transition from the unfastened state (FIG. **7**) to the fastened state (FIG. **8**), the fastener **150** is moved down toward the extended position, which causes the latch **158** to enter into the receptacle **204** and the grooves **171** of the latch **158** to receive latch elements **205** of the receptacle **204**. These latch elements **205** comprise protrusions which protrude radially inward from the receptacle **204** and extend into the grooves **171**. Then, as the fastener **150** is rotated in the first rotational direction **195**, each protrusion travels along the groove **171**, pulling the fastener **150** farther downward, until eventually the protrusion reaches a catch **172** at the end of the groove **171**. This occurs when the fastener **150** is in the first orientation. The catch **172** is designed to hold the protrusion at the end of the groove **171** unless a sufficiently large torque is applied to rotate the fastener **150** in the second rotational direction opposite the first rotational direction **195**. With the protrusion in the catch **172**, the engagement between the protrusion and groove **171** prevents the latch **158** from being withdrawn from the receptacle **204**. Thus, in this state, the fastener **150** is locked to the receptacle.

Returning to FIG. **8**, after the fastened state is achieved, the handle **152** may then be pivoted about the pivot **154** to move the handle **152** to the stowed configuration, as suggested by the dashed arrow in FIG. **8**. This results in the state illustrated in FIG. **9**. In FIG. **9**, the fastener **150** remains in the fastened state, but now the handle **152** has been stowed. This may be the state that is used during normal operation of the computing device. It may be noted that FIGS. **8** and **9** both illustrate the fastener **150** in the fastened state, but in FIG. **8** the handle **152** is deployed while in FIG. **9** the handle **152** is stowed. Thus, the state illustrated in FIG. **8** may also be referred to as a fastened and deployed state (also illustrated in FIG. **13**), while the state illustrated in FIG. **9** may be referred to as a fastened and stowed state (also illustrated in FIGS. **10** and **12**).

The transition from the fastened state to the unfastened state may occur through the reverse of the sequence of events described above. That is, starting in the state illustrated in FIG. **9** the handle **152** may first be deployed, resulting in the state illustrated in FIG. **8**. Then, the fastener **150** may be rotated in the second rotational direction, which causes the half-turn latch **158** to disengage from the latch elements **205**. The fastener **150** may also move upward (+z direction) once the half-turn latch **158** disengages, due to the spring **153**. This results in the state illustrated in FIG. **7**, in which the fastener **150** is no longer fastened. The riser cage **110** may then be removed from the computing device if desired.

As mentioned above, the fastener **150** is connected to the bracket **150** via the fastener mount **135**. As shown in FIGS. **2** and **15**, the fastener mount **135** is connected to the top portion **139** near a front end of the bracket **120** (i.e., the end opposite from the rear portion **121**, or the end in the +x direction). Note that in FIG. **15**, the fastener **150** is omitted to allow the fastener mount **135** to be better seen. As shown in FIG. **15**, the fastener mount **135** comprises vertical walls **148** and **149**, which are coupled to horizontal flanges **146** and **147**, respectively. The horizontal flanges **146** and **147** overlap with the top portion **139** and are attached thereto via fasteners **134**. The fastener mount **135** also comprises a shelf **137** with an aperture **138**. The half-turn latch **158** extends through the aperture **138** when the fastener **150** is in the extended position, as illustrated in FIG. **16** (see also FIGS.

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3, **5**, and **8**). As mentioned above, the shelf **137** may contact the system board **201** and/or receptacle **204**, which may provide physical support for riser card **110**. As shown in FIG. **15**, the fastener mount **135** also comprises a fastener **143**, which extends through the vertical wall **148** to engage with the housing **155** of the fastener **150**, thereby attaching the fastener **150** to the fastener mount **135** and hence to the bracket **120**.

In the description above, various types of electronic circuitry are described. As used herein, “electronic” is intended to be understood broadly to include all types of circuitry utilizing electricity, including digital and analog circuitry, direct current (DC) and alternating current (AC) circuitry, and circuitry for converting electricity into another form of energy and circuitry for using electricity to perform other functions. In other words, as used herein there is no distinction between “electronic” circuitry and “electrical” circuitry.

It is to be understood that both the general description and the detailed description provide examples that are explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. Various mechanical, compositional, structural, electronic, and operational changes may be made without departing from the scope of this description and the claims. In some instances, well-known circuits, structures, and techniques have not been shown or described in detail in order not to obscure the examples. Like numbers in two or more figures represent the same or similar elements.

In addition, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context indicates otherwise. Moreover, the terms “comprises”, “comprising”, “includes”, and the like specify the presence of stated features, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups. Components described as connected may be electronically or mechanically directly connected, or they may be indirectly connected via one or more intermediate components, unless specifically noted otherwise. Mathematical and geometric terms are not necessarily intended to be used in accordance with their strict definitions unless the context of the description indicates otherwise, because a person having ordinary skill in the art would understand that, for example, a substantially similar element that functions in a substantially similar way could easily fall within the scope of a descriptive term even though the term also has a strict definition.

And/or: Occasionally the phrase “and/or” is used herein in conjunction with a list of items. This phrase means that any combination of items in the list—from a single item to all of the items and any permutation in between—may be included. Thus, for example, “A, B, and/or C” means “one of {A}, {B}, {C}, {A, B}, {A, C}, {C, B}, and {A, C, B}”.

Elements and their associated aspects that are described in detail with reference to one example may, whenever practical, be included in other examples in which they are not specifically shown or described. For example, if an element is described in detail with reference to one example and is not described with reference to a second example, the element may nevertheless be claimed as included in the second example.

Unless otherwise noted herein or implied by the context, when terms of approximation such as “substantially,” “approximately,” “about,” “around,” “roughly,” and the like, are used, this should be understood as meaning that mathematical exactitude is not required and that instead a range

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of variation is being referred to that includes but is not strictly limited to the stated value, property, or relationship. In particular, in addition to any ranges explicitly stated herein (if any), the range of variation implied by the usage of such a term of approximation includes at least any inconsequential variations and also those variations that are typical in the relevant art for the type of item in question due to manufacturing or other tolerances. In any case, the range of variation may include at least values that are within $\pm 1\%$ of the stated value, property, or relationship unless indicated otherwise.

Further modifications and alternative examples will be apparent to those of ordinary skill in the art in view of the disclosure herein. For example, the devices and methods may include additional components or steps that were omitted from the diagrams and description for clarity of operation. Accordingly, this description is to be construed as illustrative only and is for the purpose of teaching those skilled in the art the general manner of carrying out the present teachings. It is to be understood that the various examples shown and described herein are to be taken as exemplary. Elements and materials, and arrangements of those elements and materials, may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features of the present teachings may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of the description herein. Changes may be made in the elements described herein without departing from the scope of the present teachings and following claims.

It is to be understood that the particular examples set forth herein are non-limiting, and modifications to structure, dimensions, materials, and methodologies may be made without departing from the scope of the present teachings.

Other examples in accordance with the present disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with the following claims being entitled to their fullest breadth, including equivalents, under the applicable law.

What is claimed is:

1. A riser cage for coupling an expansion card to a computing device, comprising:
 - a bracket configured to support the expansion card;
 - a riser card attached to the bracket and configured to electrically connect the expansion card to the computing device;
 - a fastener attached to the bracket, the fastener comprising:
 - a handle;
 - a shaft extending along a first axis;
 - a pivot connecting the handle to the shaft;
 - a half-turn latch coupled to the shaft and configured to engage with a receptacle of the computing device to selectively secure the fastener to the receptacle;
 - wherein the handle is pivotable about the pivot between a deployed configuration and a stowed configuration, and in the stowed configuration the handle is parallel to the riser card.
2. The riser cage of claim 1, wherein the shaft is rotatable about the first axis and the fastener is transitionable between a fastened state and an unfastened state by rotation of the shaft 180 degrees about the first axis.
3. The riser cage of claim 1, wherein the handle is pivotable about a second axis perpendicular to the first axis.

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4. The riser cage of claim 3, wherein the handle is configured to remain parallel to the riser card throughout movement of the handle from the stowed configuration to the deployed configuration.
5. The riser cage of claim 3, wherein in a fastened state of the fastener, the second axis is perpendicular to the riser card.
6. The riser cage of claim 1, wherein in the stowed configuration, the handle does not protrude above the bracket along a direction of the first axis.
7. The riser cage of claim 6, wherein in the deployed configuration, the handle protrudes above the bracket along the direction of the first axis.
8. The riser cage of claim 1, wherein the bracket comprises a rear portion and a transverse portion, the fastener coupled to the transverse portion; and wherein, in the stowed configuration, the handle extends in a forward direction away from the rear portion.
9. The riser cage of claim 1, wherein the handle comprises a proximal portion and a distal portion transverse to the proximal portion; wherein, in the stowed configuration, the proximal portion of the handle extends horizontally and distal portion extends vertically.
10. The riser cage of claim 9, wherein the bracket comprises a rear portion and a transverse portion, the fastener coupled to the transverse portion; and wherein in the stowed configuration, the proximal portion of the handle extends forward away from the rear portion.
11. The riser cage of claim 1, comprising the expansion card electrically connected to the riser card and attached to the bracket.
12. A computing device, comprising:
 - a chassis;
 - a primary system board comprising an expansion slot and a receptacle, the primary system board supported by the chassis;
 - a riser cage coupled to the chassis, the riser cage comprising:
 - a bracket;
 - an expansion card attached to the bracket;
 - a riser card attached to the bracket, electrically connected to the expansion card, and electrically connected to the expansion slot;
 - a fastener attached to the bracket, the fastener comprising:
 - a handle;
 - a shaft extending along a first axis;
 - a pivot connecting the handle to the shaft;
 - a half-turn latch coupled to the shaft and engaged with the receptacle to selectively secure the fastener to the receptacle;
 - wherein the handle is pivotable about the pivot between a deployed configuration and a stowed configuration, and in the stowed configuration the handle is parallel to the riser card.
13. The computing device of claim 12, wherein the shaft is rotatable about the first axis and the fastener is transitionable between a fastened state and an unfastened state by rotation of the shaft 180 degrees about the first axis.

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14. The computing device of claim 12, wherein the handle is pivotable about a second axis perpendicular to the first axis.

15. The computing device of claim 14, wherein the handle is configured to remain parallel to the riser card throughout movement of the handle from the stowed configuration to the deployed configuration. 5

16. The computing device of claim 14, wherein in a fastened state of the fastener, the second axis is perpendicular to the riser card. 10

17. The computing device of claim 12, wherein in the stowed configuration, the handle does not protrude above the bracket along a direction of the first axis. 15

18. The computing device of claim 12, wherein the bracket comprises a rear portion and a transverse portion, the fastener coupled to the transverse portion, the rear portion forming part of a rear panel of the chassis; and wherein, in the stowed configuration, the handle extends 20 in a forward direction away from the rear portion.

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19. A method, comprising:

positioning a riser cage on a primary system board of a computing device adjacent an expansion slot and a receptacle

connecting an electrical connector of a riser card of the riser cage to the expansion slot, the riser card being electrically connected to an expansion card attached to a bracket of the riser cage;

securing the riser cage to the primary system board by rotating a fastener comprising a shaft, a handle pivotably connected to the shaft, and a half-turn latch connected to the shaft, the rotation of the fastener causing the half-turn latch to engage with the receptacle to secure the fastener to the receptacle; and

pivoting the handle from a deployed configuration to a stowed configuration, wherein in the stowed configuration the handle is parallel to the riser card.

20. The method of claim 19, wherein, in pivoting the handle from the deployed configuration to the stowed configuration, the handle remains parallel to the riser card throughout the pivoting motion.

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